# THE (ALMOST) IMPOSSIBLE PUZZLE

Abstract. Tom, Jerry and Spike play the following game. Tom generates a random binary code of length $n$, which he sends to Spike along with a *secret number* between 0 and $n-1$. Spike flips exactly one bit in the code and then sends only the modified binary code to Jerry. Is there a pre-established strategy for Spike and Jerry such that Jerry can determine the *secret number* solely by looking at the code sent by Spike?

## 1. Definitions

Let $\mathrm{Fin}\, n := \{0, \ldots, n-1\}$, $n \in \mathbb{N}$ ($\mathrm{Fin}\, 0 := \emptyset$). $\mathbb{Z}_2 := \{0, 1\}$ is the commutative ring of integers modulo 2.

`def:Z2^n` **Definition 1.1.** $\mathbb{Z}_2^n := \mathbb{Z}_2 \times \ldots \times \mathbb{Z}_2$. $\mathbb{Z}_2^n$ can be seen as the set of all functions from $\mathrm{Fin}\, n$ to $\mathbb{Z}_2$. In this file, we consider $\mathbb{Z}_2^n$ as a module over $\mathbb{Z}_2$.

`def:e` **Definition 1.2.** For $i \in \mathrm{Fin}\, n$, let $\mathrm{e}_i \in \mathbb{Z}_2^n$ be the canonical vector with 1 on the $i$-th position and zeros elsewhere

`e` (e) $$\mathrm{e}_i := (0, \ldots, 0, 1, 0, \ldots, 0),$$

$\mathrm{e}_i$ can be also seen as the function defined on $\mathrm{Fin}\, n$, supported at $i$, with value 1 there, and 0 elsewhere.

Note that $b + \mathrm{e}_i$ is the code $b$ with the $i$-th bit flipped, for every $b \in \mathbb{Z}_2^n$ and $i \in \mathrm{Fin}\, n$.

`def:SF` **Definition 1.3.** We say that $f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n$ is a **strategy function**, if

`SF` (SF) $$\forall\, b \in \mathbb{Z}_2^n, k \in \mathrm{Fin}\, n, \exists\, i \in \mathrm{Fin}\, n, f(b + \mathrm{e}_i) = k.$$

We say that the puzzle **has a strategy** for $n$ if

$$\exists f : \mathbb{Z}_2^n \to \mathrm{Fin}\, n,\ (\text{SF}).$$

`def:d` **Definition 1.4.** Let $\delta : \mathrm{Fin}\, n \times \mathrm{Fin}\, n \to \mathbb{N}$ be given by

`d` (d) $$\delta(i, j) \;=\; \text{if } i = j \text{ then } 1 \text{ else } 0.$$

See the Mathlib definition if-then-else.

## 2. Main result

In this section, we characterize the values of $n$ for which the puzzle has a strategy and we present a sketch of the proof. The details of the proof are given in the next sections and are adapted for formalization in Lean.

**Theorem** *The puzzle has a strategy for $n \in \mathbb{N}^*$ if and only if $n$ is a power of* $2$.

*Sketch of the proof.*
$\boxed{\text{Modus Ponens}}$ For every $b \in \mathbb{Z}_2^n$,

$$\mathrm{Fin}\, n \ni i \mapsto f(b + \mathrm{e}_i) \in \mathrm{Fin}\, n \text{ is surjective,}$$

and thus also injective. Fix some $k \in \mathrm{Fin}\, n$ (e.g., $k = 0 < n$). Then

$$\begin{aligned} 2^n &= \sum_{b \in \mathbb{Z}_2^n} 1 = \sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \mathrm{Fin}\, n} \delta(f(b + \mathrm{e}_i), k) = \sum_{i \in \mathrm{Fin}\, n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k) \\ &= \sum_{i \in \mathrm{Fin}\, n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k). \end{aligned}$$

$\boxed{\text{Modus Ponens Reverse}}$ Let $n = 2^m$. Let $g : \mathrm{Fin}\,(2^m) \to \mathbb{Z}_2^m$ be a bijection. Define

$$f(b) := g^{-1}\left( \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) \right), \quad b \in \mathbb{Z}_2^n.$$

Let $b \in \mathbb{Z}_2^n$ and $k \in \mathrm{Fin}\, n$.

$$f(b + \mathrm{e}_i) = k \Leftrightarrow \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) + \sum_{j \in \mathrm{Fin}\, 2^m} \mathrm{e}_{ij} \cdot g(j) = g(k)$$

$$\Leftrightarrow \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) + g(i) = g(k) \Leftrightarrow i = g^{-1}\left( g(k) - \sum_{j \in \mathrm{Fin}\, 2^m} b_j \cdot g(j) \right).$$

## 3. Lean tactics

- apply
- by_contra
- constructor
- dsimp
- exact
- have
- intro
- let
- obtain
- rw
- simp (only)
- unfold
- use

## 4. Function shift is injective

fun_shift_surj

**Lemma 4.1.** *If $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ is (SF) and $b \in \mathbb{Z}_2^n$, then the function $\operatorname{Fin} n \ni i \mapsto f(b + \mathrm{e}_i) \in \operatorname{Fin} n$ is surjective.*

*Proof.* $\operatorname{Fin} n \ni i \mapsto f(b + \mathrm{e}_i) \in \operatorname{Fin} n$ is surjective, if

$$\forall k \in \operatorname{Fin} n, \exists i \in \operatorname{Fin} n, f(b + \mathrm{e}_i) = k.$$

Taking our $b$ in (SF), we get the desired conclusion. □

`Mathlib hint:` Function.Surjective.

fun_shift_inj

**Lemma 4.2.** *If $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ is (SF) and $b \in \mathbb{Z}_2^n$, then the function $\operatorname{Fin} n \ni i \mapsto f(b + \mathrm{e}_i) \in \operatorname{Fin} n$ is injective.*

*Proof.* Since $\operatorname{Fin} n \ni i \mapsto f(b+\mathrm{e}_i) \in \operatorname{Fin} n$ is a surjective function, by Lemma 4.1, on a finite set, it is also injective. □

`Mathlib hints:` Function.Injective, Finite.injective_iff_surjective.

## 5. Function shift sum

fun_shift_sum_one

**Lemma 5.1.** *If $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ is (SF), $b \in \mathbb{Z}_2^n$, and $k \in \operatorname{Fin} n$, then*

$$\sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k) = 1.$$

*Proof.* Since $f$ is (SF), we obtain $i_k \in \operatorname{Fin} n$ such that $f(b + \mathrm{e}_{i_k}) = k$.

We shall rewrite the sum by proving that only the $i_k$ term in the sum is equal to 1, and the rest are 0.

Clearly, $\delta(f(b + \mathrm{e}_{i_k}), k) = 1$, in view of (d).

Introduce $i \in \operatorname{Fin} n$ with $i \neq i_k$. Rewritting $k$ as $f(b+\mathrm{e}_{i_k})$, $\delta(f(b+\mathrm{e}_i), k) = 0$, if we prove that $f(b + \mathrm{e}_i) \neq f(b + \mathrm{e}_{i_k})$. In order to get a contradiction, assume that $f(b + \mathrm{e}_i) = f(b + \mathrm{e}_{i_k})$. Then, by Lemma 4.2, we have $i = i_k$, which is false in view of the assumption $i \neq i_k$. □

`Mathlib hints:` rw [Fintype.sum_eq_single $i_k$], if_pos, if_neg.

fun_shift_sum_pow2

**Lemma 5.2.** *If $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ is (SF) and $k \in \operatorname{Fin} n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k) = 2^n.$$

*Proof.* By Lemma 5.1 for $f$ with the hypothesis (SF), we have:

$$\forall b \in \mathbb{Z}_2^n, \sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k) = 1.$$

With only this relation, we can simplify $\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k)$ to $\sum_{b \in \mathbb{Z}_2^n} 1$, which simplifies further to $2^n$. □

fun_shift_sum_rev

**Lemma 5.3.** *If $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ and $k, i \in \operatorname{Fin} n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k) = \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k).$$

*Proof.* Let $g : \mathbb{Z}_2^n \to \mathbb{Z}_2^n, g(b) = b + \mathrm{e}_i$. Then $g$ is bijective. Applying the fact that $g$ is bijective to our finite sum, the equality holds, because $\sum_{b \in \mathbb{Z}_2^n} \delta(f(g(b)), k) = \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k)$. □

`Mathlib hints:` Function.Bijective, add_left_injective, add_right_surjective, Fintype.sum_bijective.

## 6. Modus Ponens

strategy_pow2

**Lemma 6.1.** *If $0 < n$ and $f : \mathbb{Z}_2^n \to \operatorname{Fin} n$ is (SF), then $\exists N \in \mathbb{N}, n \cdot N = 2^n$.*

*Proof.* Let $k := 0$ (which is in $\operatorname{Fin} n$ since $0 < n$). By Lemma 5.2, we have

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \operatorname{Fin} n} \delta(f(b + \mathrm{e}_i), k) = 2^n.$$

Since the sums commute, we can rewrite the above expression as

$$\sum_{i \in \operatorname{Fin} n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + \mathrm{e}_i), k) = 2^n.$$

Using Lemma 5.3, we simplify this to have

$$\sum_{i \in \operatorname{Fin} n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = 2^n.$$

We use $N := \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k)$ to finish the proof. □

`Mathlib hint:` Finset.sum_comm.

mul_pow2

**Lemma 6.2.** *If $\exists N \in \mathbb{N}, n \cdot N = 2^n$, then $\exists m \in \mathbb{N}, n = 2^m$.*

*Proof.* From the hypothesis, we obtain $N$ such that $n \cdot N = 2^n$. So, we have $n$ divides $2^n$. Since 2 is prime, $\exists\, m \in \mathbb{N}, n = 2^m$. □

`Mathlib hints:` Dvd.intro, Nat.dvd_prime_pow.

puzzle_mp

**Theorem 6.3.** *If $0 < n$ and the puzzle has a strategy for $n$, then $\exists\, m \in \mathbb{N}, n = 2^m$.*

*Proof.* Since the puzzle has a strategy, we obtain $f$ with (SF). By putting toghether Lemma 6.1 and Lemma 6.2, $\exists\, m \in \mathbb{N}, n = 2^m$. □

## 7. Modus Ponens Reverse

`bij_Fin` **Definition 7.1.** For $m \in \mathbb{N}$, let $g : \mathrm{Fin}\,(2^m) \to \mathbb{Z}_2^m$ be a bijective function. $g$ exists, applying the fact that $\mathrm{Fin}\,(2^m)$ and $\mathbb{Z}_2^m$ have equal cardinals.

`Mathlib hint:` Fintype.equivOfCardEq.

`sum_basis` **Lemma 7.2.** *If $i \in \mathrm{Fin}\,n$ and $f : \mathrm{Fin}\,n \to \mathbb{Z}_2^m$, then* $\sum_{j \in \mathrm{Fin}\,n} e_{ij} \cdot f(j) = f(i)$.

*Proof.* Unfolding (e), $e_{ij} = 1$ if $i = j$ and $e_{ij} = 0$ otherwise. We shall rewrite $\sum_{j \in \mathrm{Fin}\,n} e_{ij} \cdot f(j)$ as $f(i)$, using the fact that the single term that contributes in the sum is the one with $j = i$. □

`Mathlib hint:` rw[Fintype.sum_single_smul].

`puzzle_mpr` **Theorem 7.3.** *If $n = 2^m$ for some $m$, then the puzzle has a strategy for $n$.*

*Proof.* First, we rewrite $n$ as $2^m$ in (SF). Let $g$ be given by Definition 7.1 and use $f : \mathbb{Z}_2^n \to \mathrm{Fin}\,n$ given by

$$f(b) := g^{-1}\left(\sum_{j \in \mathrm{Fin}\,2^m} b_j \cdot g(j)\right), \quad b \in \mathbb{Z}_2^n.$$

Introduce $b \in \mathbb{Z}_2^n$ and $k \in \mathrm{Fin}\,n$ in order to verify (SF) for $f$. In view of (SF), we want to show

$$\exists\, i \in \mathrm{Fin}\,n, f(b + \mathrm{e}_i) = g^{-1}\left(\sum_{j \in \mathrm{Fin}\,2^m} (b_j + e_{ij}) \cdot g(j)\right) = k.$$

We simplify the above relation using the distributivity of the scalar multiplication over addition and over sum to get

$$\sum_{j \in \mathrm{Fin}\,2^m} (b_j + e_{ij}) \cdot g(j) = \sum_{j \in \mathrm{Fin}\,2^m} b_j \cdot g(j) + \sum_{j \in \mathrm{Fin}\,2^m} e_{ij} \cdot g(j).$$

Let's use $i := g^{-1}\left(g(k) - \sum_{j \in \mathrm{Fin}\,2^m} b_j \cdot g(j)\right)$. By Lemma 7.2, we can rewrite the desired relation with $\sum_{j \in \mathrm{Fin}\,2^m} e_{ij} \cdot g(j) = g(i)$. By the expression of $i$, we can simplify to get

$$\begin{aligned} &g^{-1}\left(\sum_{j \in \mathrm{Fin}\,2^m} b_j \cdot g(j) + g(i)\right) \\ &= g^{-1}\left(\sum_{j \in \mathrm{Fin}\,2^m} b_j \cdot g(j) + g(k) - \sum_{j \in \mathrm{Fin}\,2^m} b_j \cdot g(j)\right) = g^{-1}(g(k)) = k. \end{aligned}$$

□

`Mathlib hints:` g.symm, add_smul, Finset.sum_add_distrib.

## 8. Finish

puzzle

**Theorem 8.1.** *If* $0 < n$, *then the puzzle has a strategy for* $n$ *if and only if* $\exists\, m \in \mathbb{N}, n = 2^m$.

*Proof.* The forward direction is Theorem 6.3, and the reverse direction is Theorem 7.3. $\square$